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GRAPH NEURAL NETWORKS FOR AUTONOMOUS THEOREM PROVING

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Contents

Abstract **ii**

1 Introduction **1**

2 Simple Type Theory **3**

3 Generalized Algebraic Theories **5**
3.1 Formal Definition 7
3.2 The Substitution Lemma 9
3.3 Models and Homomorphisms 11
3.4 Contexts and Realizations 13

4 Calculus of Constructions **19**

5 Expansion Algorithm **21**
5.1 Problem Definition 21
5.2 Expansion algorithm 22

Bibliography **29**

Abstract

The main goal of this work is to prove the existence of.....

Notation: We will work over an algebraically closed field K of characteristic zero.

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Chapter 1

Introduction

Introduction

Chapter 2

Simple Type Theory

Chapter 3

Generalized Algebraic Theories

The goal of this chapter is give meaning to the concept of term, type and theory, while introducing some tools that we will use later when we present the algorithm.

By theories we mean Generalized Algebraic Theories (or GATs), introduced by John Cartmell in his thesis [1] [2]. GATs give us both a very idiomatic way of presenting theories and powerful interpretation tools like Context Categories, introduced later in this section. The notion of generalized algebraic theory is a generalization of the notion of many-sorted algebraic theory in the following manner: while in Martin Löff's many-sorted algebraic theories sorts are constant types, interpreted as sets, in generalized algebraic theories sorts might be constant or variable, over a specified range of types, and variable types are interpreted as families of sets. So a generalized algebraic theory consists of:

1. a set of sorts, that can be constant or variable;
2. a set of operators, with the sort of the arguments and value specified;
3. a set of axioms, where each axiom is an identity between similar expressions.

Let's present category theory for example: The set of sorts contains two elements Ob and Hom. Ob is a constant type and Hom is a variable type, for a pair of terms of type Ob, name them x, y, we have a type $Hom(x, y)$. Here we see a key point in general algebraic theories, types vary over terms, no over types. The set of operators also has two elements id and $\circ$. The id operator has one argument x of type Ob, and returns a value of type $Hom(x, x)$. This shows how the type of value of an operator might vary depending on the arguments. The $\circ$ operator (or composition) expects 5 arguments, three of type Ob, name them x, y, z, one of type $Hom(x, y)$, say f, and one of type $Hom(y, z)$, named g, and it returns a value of type $Hom(x, z)$. Here we see that the type of an argument might also vary

depending on a previous argument. Informally, instead of writing $\circ(x, y, z, f, g)$ we can write $\circ(f, g)$, as the first three arguments might be inferred from the last two. This notation will be discussed in a further section.

Now we will introduce the notation we will be using when talking about terms and types. This notation will be formally defined in a later section. So consider you have a set of variables V, we will associate terms to variables as required. When asserting that a term t is of type Δ we will write $t : \Delta$, but if t has variables $x_1, \ldots, x_n$ occurring in it this assertion does not make sense unless we assign some types to $x_1, \ldots, x_m$. To do this we will write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

that reads as: if x_1 is a variable of type Δ_1, ...and if x_n is a variable of type Δ_n, then t is a term of type Δ. Similarly, if we want to assert that if x_1 is a variable of type Δ_1, ...and if x_n is a variable of type Δ_n, then Δ is a type, we can write:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

this assertions are called rules. Rules are used to express which expressions are well-formed. We will work with this rules as units instead of the base expressions. This is because terms and types do not make sense by themselves but they need to be thought of in a context.

Axioms are also written as rules. The conclusion of those rules are equalities between types or between terms of the same type. We write those rules as:

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'} \qquad \frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}$$

and read as: if x_1 is a variable of type Δ_1, ...and x_n is a variable of type Δ_n, then $\Delta = \Delta'$, or $t = t'$ as terms of type Δ, respectively.

To continue with the previous example, we can write the axioms of category theory as:

$$\begin{aligned}
&x, y : Ob, f : Hom(x, y), \quad \circ(id(x), f) = f \\
&x, y : Ob, f : Hom(x, y), \quad \circ(f, id(y)) = f \\
&w, x, y, z : Ob, f : Hom(w, x), g : Hom(x, y), h : Hom(y, z), \\
&\qquad \circ(\circ(f, g), h) = \circ(f, \circ(g, h))
\end{aligned}$$

When presenting a theory's language i.e. the sets of sorts and operators, we must present each symbol with a rule specifying the role of this symbol, either as a sort symbol or as specifically typed operator symbol. This rules are called introductory rules. In the case of a sort symbol A the rule should be like

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, x_n) \text{ is a type}}$$

clearly specifying the types A is dependent on. In the case of an operator symbol, a rule of the form

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{f(x_1, \ldots, x_n) : \Delta}$$

suffices to specify the types of the arguments and the value of f.

Finally, to present a theory we will give a set os symbols, each associated to its introductory rule, and a set of axioms. And, of course, everything must be well-formed, something we will address in the following section. So now we can give a formal presentation of the theory of categories:

Symbol	*Introductory Rule*
Ob	Ob is a type.
Hom	$x, y \in \mathrm{Ob} : \mathrm{Hom}(x, y)$ is a type.
$\circ$	$x, y, z \in \mathrm{Ob}$, $f \in \mathrm{Hom}(x, y)$, $g \in \mathrm{Hom}(y, z) : \circ(f, g) \in \mathrm{Hom}(x, z)$.
id	$x \in \mathrm{Ob} : \mathrm{id}(x) \in \mathrm{Hom}(x, x)$.

Axioms

$\circ(\mathrm{id}(x), f) = f$,	whenever $x, y \in \mathrm{Ob}$ and $f \in \mathrm{Hom}(x, y)$.
$\circ(f, \mathrm{id}(y)) = f$,	whenever $x, y \in \mathrm{Ob}$ and $f \in \mathrm{Hom}(x, y)$.
$\circ(\circ(f, g), h) = \circ(f, \circ(g, h))$,	whenever $w, x, y, z \in \mathrm{Ob}$, $f \in \mathrm{Hom}(w, x)$, $g \in \mathrm{Hom}(x, y)$ and $h \in \mathrm{Hom}(y, z)$.

3.1 Formal Definition

In this section we will formally define all of the concepts we just introduced. To give a notion of well-formed introductory rule, we need the concept of drivability, which also depends on the introductory rules. To tackle this problem we will leave aside the question of well-formedness until we have the complete set of derived rules of a theory. We will first define a pretheory and then define a theory as a well-formed pretheory.

We will assume to have countable set of variables V. Lets define the notion of rule on an alphabet W, the elements of W are called symbols.

Definition 3.1. *The set of* ***expressions*** *is defined inductively such that expressions are finite sequences of elements of $W \cup V \cup \{(\} \cup \{)\} \cup \{,\}$ defined by the clauses:*

1. *if $x \in V$, then x is an expression;*
2. *if $L \in W$, then L is an expression;*
3. *if $L \in W$ and $e_1, \ldots, e_n$ are expressions, then $L(e_1, \ldots, e_n)$ is an expression.*

Now we can define the building blocks of a rule:

Definition 3.2. *A **premise** is defined to be a sequence of $V\times$ the set of expressions. The empty sequence is allowed and we call it an empty premise. If $((x_1, \Delta_1), \ldots, (x_n, \Delta_n))$ is a premise, we write it as $x_1 : \Delta_1, \ldots, x_n : \Delta_n$.*

Definition 3.3. *We call a **conclusion** to one of the following:*

1. *T-conclusion: defined by a single expression Δ and written as 'Δ is a type'.*
2. *$\in$-conclusion: defined by a pair of expressions (t, Δ) and written as '$t \in \Delta$'.*
3. *T =conclusion: defined by a pair of expressions (Δ_1, Δ_2) and written as '$\Delta_1 = \Delta_2$'.*
4. *$\in$=conclusion: defined by a triple of expressions (t_1, t_2, Δ) and written as '$t_1 = t_2 : \Delta$'*

Finally a **rule** is determined by a premise P and a conclusion C, and is written as $\frac{P}{C}$. We name the rule as T-rule, $\in$-rule, T =rule or $\in$=rule according to the form of the conclusion.

Now we can define pretheory and later theory.

Definition 3.4. *A **pretheory** consists of*

1. *a set of sort symbols S;*
2. *a set of operator symbols Z;*
3. *for each sort symbol A, an associated rule of the alphabet $S \cup Z$:*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{A(x_1, \ldots, A_n) \text{ is a type}}$$

   for some $n \geq 0$. It is called the introductory rule of A;

4. *for each operator symbol F, an associated rule of the alphabet $S \cup Z$:*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{F(x_1, \ldots, x_n) : \Delta}$$

   for some $n \geq 0$. It is called the introductory rule of F;

5. *a set of axioms. Each axiom is either a T =rule or a $\in$=rule of the alphabet $S \cup Z$.*

Now we will tackle the problem of well-formedness of rules.

Definition 3.5. *If U is a pretheory, then*

1. *a* ***context*** *is a premise* $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ *such that the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{ is a type}}$$

is a derived rule such that x_n *is distinct from all of* $x_1, \ldots, x_n$.

2. (a) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ *is* ***well-formed*** *if the premise* $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ *is a context.*

   (b) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ *is* ***well-formed*** *if* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ *is a derived rule of* U.

   (c) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$ *is* ***well-formed*** *if* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ *and* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta' \text{ is a type}}$ *are derived rules of* U.

   (d) *the rule* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = t' : \Delta}$ *is* ***well-formed*** *if* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$ *and* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t' : \Delta}$ *are derived rules of* U.

Definition 3.6. *The set of* ***derived rules*** *of a pretheory* U *is the set of rules derivable by the following principles of derivation:*

Definition 3.7. *A pretheory is well-formed if all of its introductory rules are well-formed. A* ***theory*** *is a well-formed pretheory.*

3.2 The Substitution Lemma

Given $\Delta, t_1, \ldots, t_n$ expressions and $x_1, \ldots, x_n$ variables, then we define the expression $\Delta[t_1|x_1, \ldots, t_n|x_n]$ the result of simultaneous replacement of $x_1, \ldots, x_n$ in Δ by $t_1, \ldots, t_n$. We call this the Substitution operation.

The substitution lemma states that the set of derived rules of a theory is closed under the operation of substitution of correctly typed terms for variables. We could add substitution as one of the derivation rules but would make applying induction very messy.

Definition 3.8. *A derived rule of* $\dfrac{y_1|\Omega_1, \ldots, y_n|\Omega_n}{Conclusion}$ *of a theory* U *is said to have the* ***substitution property*** *if for all contexts* Q *of* U*, for all expressions* $s_1, \ldots, s_m$ *of* U *if the rules*

$$\frac{Q}{s_1 : \Omega_1}, \frac{Q}{s_2 : \Omega_2[s_1|y_1]}, \ldots, \frac{Q}{s_m : \Omega_m[s_1|y_1, \ldots, s_{m-1}|y_{m-1}]}$$

are all derived rules of U, then the rule

$$\frac{Q}{Conclusion[s_1|y_1, \ldots, s_m|y_m]}$$

is a derived rule of U.

We will see that all derived rules hold this property.

Lemma 3.9. *If* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{Conclusion}$ *is a derived rule of U, then any variable appearing in the conclusion occurs among $\{x_1, \ldots, x_n\}$.*

Proof. Easily obtained by applying induction over the derivation rules. □

Lemma 3.10.

1. *the premise of a derived rule is a context,*
2. *if $x_1 : \Delta_1, \ldots, x_n : \Delta_n$ is a context, then for all $1 \leq i \leq n$, then* $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i \text{ is a type}}$ *is a derived rule.*

Proof. □

Lemma 3.11 (The Substitution Lemma)**.** *Every derived rule of a theory U has the substitution property.*

Proof. □

Corollary 3.12 (Change of Variables)**.** *If* $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{Conclusion}$ *is a derived rule of U and if $y_1, \ldots, y_n$ is a sequence of distinct variables, then*

$$\frac{y_1 : \Delta_1, y_2 : \Delta_2[y_1|x_1], \ldots, y_n : \Delta_n[y_1|x_1, \ldots, y_{n-1}|x_{n-1}]}{Conclusion[y_1|x_1, \ldots, y_n|x_n]}$$

is a derived rule of U.

3.3 Models and Homomorphisms

In this section we will consider the semantics of a theory. To do this we will introduce the notion of model model homomorphism.

First, we will set some notation and basic concepts. We will consider a **tree** to be a tree with a unique least element, in the case that our tree does not have one we will add it. This element will be denoted as 1. Let Θ be a tree. To denote that A is a node of Θ we will write $A \in \Theta$, confusing the tree with its set of nodes. To assert that B is a node of Θ and B succeeds A, we write $A \lhd B$ in Θ. Also, if A is a node of Θ and $A \neq 1$, there exists a unique $n \geq 0$ and $A_1, \ldots, A_n$ such that $1 \lhd A_1 \lhd \cdots \lhd A_n \lhd A$ in Θ.

The set of contexts of a theory U is structured as a tree. Where its least element is the empty context and for any $n \geq 1$ the predecessor of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is $\langle x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1} \rangle$.

We want to consider the tree of sets, families of sets, families of families of sets, ... Consider the following notation for families: given A is a set, if for every element $a \in A$ $B(a)$ is a set, then the corresponding A-indexed family of sets is denoted as $\lambda a \in A.B(a)$ or as $\lambda a.B(a)$; If we have a set $C(a,b)$ for every $a \in A$ and $b \in B(a)$, then $\lambda a \in A.\lambda b \in B(a).C(a,b)$ is an A-indexed family of families of sets; and so on. The collection of sets, families of sets, families of families of sets, and so on, is called the **tree of families**. Given $1 \lhd A_1 \lhd A_2 \lhd \cdots \lhd A_n \lhd A$ in the tree of families. Then, for all $a_1 \in A_1$, ..., for all $a_n \in A_n(a_1, \ldots, a_{n-1})$, $A(a_1, \ldots, a_n)$ is a set. Finally we should define the concept of **operator**. If we have $1 \lhd A_1 \lhd \cdots \lhd A_n \lhd A$ in the tree of families. If for all $a_1 \in A_1$, ..., for all $a_n \in A_n(a_1, \ldots, a_{n-1})$, $f(a_1, \ldots, a_n) \in A(a_1, \ldots, a_n)$. We say that $\lambda a_1 \in A_1 \ldots \lambda a_n \in A_n(a_1, \ldots, a_{n-1}).f(a_1, \ldots, a_n)$ is an operator at A.

Now we will introduce the notion of a model of a theory. It is important to understand that the derived rules are the ones to be interpreted by the model, not expressions. To denote the interpretation of a rule R by a model M we will use the superscripted R^M.

Definition 3.13. *Let U be a theory. M is a model of U if*

1. *for each derived T-rule R of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$, if we consider the rules R_i to be $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i \text{ is a type}}$ for all $1 \leq i \leq n$. Then M assigns those rules to $1 \lhd R_1^M \lhd R_2^M \lhd \cdots \lhd R_n^M \lhd R^M$ in the tree of families;*

2. *if R_t is a derived $\in$-rule of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$, following the notation of the previous point, M assigns the rule to R_t^M an operator at R^M.*

Definition 3.14. *If M and M' are models of U, $f : M \to M'$ is a* ***homomorphism*** *if*

1. *for every derived T-rule R of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ there is an operator f_R such that for every $a_1 \in R_1^M$, ..., for every $a_n \in R_n^M(a_1, \ldots, a_{n-1})$ and for every $a \in R^M(a_1, \ldots, a_n)$,*

$$f_R(a_1, \ldots, a_n, R_t^M(a_1, \ldots, a_n)) \in R^{M'}(f_{R_1}(a_1), \ldots, f_{R_n}(a_1, \ldots, a_n))$$

2. *for all derived $\in$-rules R_t of the form $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$, for all $a_1 \in R_1^M, \ldots, a_n \in R_n^M(a_1, \ldots, a_{n-1})$,*

$$f_R(a_1, \ldots, a_n, R_t^M(a_1, \ldots, a_n)) = R_t^{M'}(f_{R_1}(a_1), \ldots, f_{R_n}(a_1, \ldots, a_n))$$

3. *if $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$ is a derived rule of U, then $f_R = f_{R'}$, where R and R' are the rules $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$ and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta' \text{ is a type}}$ respectively.*

To define a homomorphism it is enough to give a family $\lambda A \in S.f_A$ of correctly typed operators, where S is the set of sort symbols of U.

The category of models and homomorphisms of a theory U is named $U - alg$.

We want to create an equivalence relation on the derived T- and $\in$-rules of a theory U which we call the **equivalence relation of identity of denotation** such that $R \equiv R'$ by this relation if any model should interpret R and R' as identical objects.

Example 3.15. Semantically we should not distinguish between the rules

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}} \qquad \frac{x'_1 : \Delta_1, \ldots, x'_n : \Delta_n[x'_1|x_1, \ldots, x'_{n-1}|x_{n-1}]}{\Delta[x'_1|x_1, \ldots, x'_n|x_n] \text{ is a type}}$$

that only differ on the choice of the variables.

Example 3.16. We should not distinguish either between

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}} \qquad \frac{x_1 : \Delta'_1, \ldots, x_n : \Delta'_n}{\Delta' \text{ is a type}}$$

if for each i, $1 \leq i \leq n$, $\dfrac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i = \Delta'_i}$ and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta = \Delta'}$.

We define the relation $\equiv$:

- between the T-rules of a theory U by

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta_{n+1} \text{ is a type}} \equiv \frac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{\Omega_{m+1} \text{ is a type}}$$

if, and only if, $m = n$ and for each $1 \leq i \leq n+1$

$$\frac{x_1 : \Delta_1, \ldots, x_{i-1} : \Delta_{i-1}}{\Delta_i = \Omega_i[x_1|y_1, \ldots, x_{i-1}|y_{i-1}]}$$

is a derived rule of U.

- between $\in$-rules by

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta} \equiv \frac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{s : \Omega}$$

if, and only if,

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}} \equiv \frac{y_1 : \Omega_1, \ldots, y_m : \Omega_m}{\Omega \text{ is a type}}$$

and $\dfrac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t = s[x_1|y_1, \ldots, x_n|y_n] : \Delta}$ is a derived rule of U.

3.4 Contexts and Realizations

In this section we will introduce context categories, a powerful interpretation of general algebraic theories that will be key later on. We will start by defining the category of contexts and realizations of a theory U.

Definition 3.17. *If $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ and $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ are contexts of U, then a **realization** of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is a m-tuple $\langle t_1, \ldots, t_m \rangle$ such that for each j, $1 \leq j \leq m$, the rule*

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t_j : \Omega_j[t_1|y_1, \ldots, t_{j-1}|y_{j-1}]}$$

is a derived rule of U.

Then, for a theory U, we define the category of contexts and realizations of U, $R(U)$, as the category whose objects are contexts of U, and where we have a morphism

$$\langle t_1, \ldots, t_m \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \longrightarrow \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$$

for each realization $\langle t_1, \ldots, t_m \rangle$ of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$. The identity of an object $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ is the realization $\langle x_1, \ldots, x_n \rangle$. And the composition of two morphisms

$$\langle t_1, \ldots, t_m \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \longrightarrow \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$$

$$\langle s_1, \ldots, s_w \rangle : \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle \longrightarrow \langle z_1 : \Lambda_1, \ldots, z_w : \lambda_w \rangle$$

is defined as

$$\langle s_1, \ldots, s_w \rangle \circ \langle t_1, \ldots, t_m \rangle = \langle s_1[t_1|y_1, \ldots, t_m|y_m], \ldots, s_w[t_1|y_1, \ldots, t_m|y_m] \rangle$$

The set of objects of $R(U)$ i.e. the set of contexts of U is structured as a tree with a single least element $\langle\rangle$, the empty context, and where every context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n$ is preceded by $x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}$. Moreover, for any context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x_{n+1} : \Delta_{n+1}, \ldots, x_{n+m} : \Delta_{n+m} \rangle$ of U, $\langle x_1, \ldots, x_n \rangle$ is a realization of $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ with respect to $\langle x_1 : \Delta_1, \ldots, x_{n+m} : \Delta_{n+m} \rangle$. We name that morphism as $p(\langle x_1 : \Delta_1, \ldots, x_{n+m} : \Delta_{n+m} \rangle, \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle)$.

Lemma 3.18. *If $A = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, $A' = \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ and $B = \langle y_1 : \Omega_1, \ldots, y_{m+w} : \Omega_{m+w}$ are contexts, and $f = \langle t_1, \ldots, t_m \rangle$ is a realization of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard to $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, then*

$$\begin{tikzcd} f^*B \arrow[r, "q(f,B)"] \arrow[d, "p(f^*B,A)"'] & B \arrow[d, "p(B,A')"] \\ A \arrow[r, "f"'] & A' \end{tikzcd}$$

is a pullback diagram in $R(U)$, where

$$f^*B = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_{m+1} : \Omega_{m+1}[t_1|y_1, \ldots, t_m|y_m], \\ \ldots, y_{m+w} : \Omega_{m+w}[t_1|y_1, \ldots, t_m|y_m] \rangle$$

$$q(f, B) = \langle t_1, \ldots, t_m, y_{m+1}, \ldots, y_{m+w} \rangle$$

This properties motivate the following definition:

Definition 3.19. *A **contextual category** consists of*

1. *a category C with a terminal object 1;*
2. *a tree structure on the objects of C such that the terminal object 1 is the unique least element of the tree;*
3. *for all $A, B \in ObC$ such that $A \lhd B$ a morphism $p(B) : B \to A$ in C and that will be written as $B \twoheadrightarrow A$;*

4. *for all $A, A' \in ObC$ and for all $f : A \to A'$ in C, for all $B \in ObC$ such that $A' \lhd B$, an object f^*B in C and a morphism $q(f, B) : f^*B \to B$ such that the diagram*

$$\begin{tikzcd} f^*B \arrow[r, "q(f,B)"] \arrow[d] & B \arrow[d] \\ A \arrow[r, "f"'] & A' \end{tikzcd}$$

is a pullback in C such that

   (a) *for all $A, B \in ObC$ such that $A \lhd B$, $id_A^*B = B$ and $q(id_A, B) = id_B$;*

   (b) *when*

$$\begin{tikzcd} & & B \arrow[d] \\ A \arrow[r, "f"] & A' \arrow[r, "f'"] & A'' \end{tikzcd}$$

   in C, then $(f' \circ f)^*B = f^*(f'^*B)$ and $q(f' \circ f, B) = q(f', B) \circ q(f, f'^*B)$.

The category $R(U)$ is structured as a contextual category by the tree structure and pullbacks defined earlier.

The contextual category $R(U)$ has an homomorphic image $C(U)$ induced by the equivalence relation of intended equivalence of denotation. More precisely, $C(U)$ is the category of equivalence classes of contexts and realizations under the following relation:

- Among contexts:

$$\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \equiv \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$$

  if, and only if,

$$\frac{x_1 : \Delta_1, \ldots, x_{n-1} : \Delta_{n-1}}{\Delta_n \text{ is a type}} \equiv \frac{y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{\Omega_m \text{ is a type}}$$

- If we have two realizations $\lambda t_1, \ldots, t_m \rangle$ of $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ with regard to $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, and $\lambda s_1, \ldots, s_m \rangle$ of $\langle y'_1 : \Omega'_1, \ldots, y'_m : \Omega'_m \rangle$ with regard to $\langle x'_1 : \Delta'_1, \ldots, x'_n : \Omega'_n \rangle$, we say that $\langle t_1, \ldots, t_m \rangle \equiv \langle s_1, \ldots, s_m \rangle$ if, and only if, for each $1 \leq j \leq m$,

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t_j : \Omega_j[t_1|y_1, \ldots, t_{j-1}|y_{j-1}]} \equiv \frac{x'_1 : \Delta'_1, \ldots, x'_n : \Delta'_n}{s_j : \Omega'_j[s_1|y'_1, \ldots, s_{j-1}|y'_{j-1}]}$$

We will define *Fam*, a category that plays the same role among contextual categories as *Set* among categories. The tree of objects of *Fam* is the tree of sets, families of sets, families of families of sets and so on, with a formally adjoined least element 1. For $n, m \geq 0$, if $1 \lhd A_1 \lhd \cdots \lhd A_n$ and $1 \lhd B_1 \lhd \cdots \lhd B_m$ in *Fam*, then the set of morphisms of A_n into B_n, is the set of m-tuples $\langle f_1, \ldots, f_m \rangle$ where $f_1, \ldots, f_m$ are operators such that for each $1 \leq j \leq m$, for $a_1 : A_1, \ldots, a_n : A_n(a_1, \ldots, a_{n-1})$, $f_j(a_1, \ldots, a_n) \in B_j(f_1(a_1, \ldots, a_n), \ldots, f_{j-1}(a_1, \ldots, a_n))$. For $n = 0$, $Hom_{Fam}(1, B_m) = \{\langle b_1, \ldots, b_m \rangle | b_1 \in B_1, \ldots, b_m \in B_m(b_1, \ldots, b_{m-1})\}$. And for $m = 0$, $Hom_{Fam}(A_n, 1) = \{\langle\rangle\}$, hence 1 is the terminal object of *Fam*. To define composition in *Fam* let $1 \lhd A_1 \lhd \cdots \lhd A_n$, $1 \lhd B_1 \lhd \cdots \lhd B_m$, $1 \lhd C_1 \lhd \cdots \lhd C_w$, $\langle f_1, \ldots, f_m \rangle : A_n \to B_m$ and $\langle g_1, \ldots, g_w \rangle : B_m \to C_w$, then $\langle g_1, \ldots, g_w \rangle \circ \langle f_1, \ldots, f_m \rangle = \langle h_1, \ldots, h_w \rangle$ where for all $1 \leq k \leq w$, h_k is defined as

$$h_k(a_1, \ldots, a_n) = g_k(f_1(a_1, \ldots, a_n), \ldots, f_m(a_1, \ldots, a_n))$$

And to define the projections, we consider $1 \lhd A_1 \lhd \cdots \lhd A_n \lhd A$ in *Fam*, then we consider $p(A) : A \to A_n$ in *Fam* given by

$$p(A) = \langle h_1, \ldots, h_n \rangle; \quad h_i(a_1, \ldots, a_n, a) = a_i$$

Finally, we will define the pullbacks. Let $1 \lhd A_1 \lhd \cdots \lhd A_n$, $1 \lhd B_1 \lhd \cdots \lhd B_m \lhd B$ in *Fam* and $\langle f_1, \ldots, f_m \rangle : A_n \to B_m$ in *Fam*, then we define $\langle f_1, \ldots, f_m \rangle^* B$ as

$$\lambda a_1 \in A_1 . \lambda a_2 \in A_2(a_1) \ldots . \lambda a_n \in A_n(a_1, \ldots, a_{n-1}). \\ B(f_1(a_1, \ldots, a_n), \ldots, f_m(a_1, \ldots, a_n))$$

and $q(\langle f_1, \ldots, f_m \rangle, B) = \langle f_1, \ldots, f_m, \gamma \rangle$ where γ is the operator defined as $\gamma(a_1, \ldots, a_n, b) = b$.

Homomorphisms between contextual categories are called contextual functors:

Definition 3.20. *If C and C' are contextual categories, then a functor $F : C \to C'$ is a* ***contextual functor*** *if*

1. *$F(1) = 1$ and if $A \lhd B$ in C, then $F(A) \lhd F(B)$ in C';*
2. *for all $A \in ObC$, $F(p(A)) = p(F(A))$;*
3. *for all f and B such that f^*B is defined in C, $F(f^*B) = F(f)^*F(B)$ and $F(q(f, B)) = q(F(f), F(B))$.*

The category of contextual category and contextual functors is named *Con*.

Now we will introduce some final definitions and results regarding contextual categories. Consider $A \leq B$ in a contextual category C, then we define $p(B, A)$:

$B \to A$ in C, written as $B \rightarrowtriangle A$, by $p(B, A) = p(X_1) \circ \cdots \circ p(X_n) \circ p(B)$ where $A \lhd X_1 \lhd \cdots \lhd X_n \lhd B$ in C. Analogously, whenever $A' \leq B$ in C and $f : A \to A'$ in C, we have the following pullback:

$$\begin{tikzcd}
q(\ldots q(f, X_1) \ldots, X_n)^* B \arrow[r, "q(q(\ldots q(f,X_1)\ldots,X_n),B)"] \arrow[d] & B \arrow[d] \\
A \arrow[r, "f"'] & A'
\end{tikzcd}$$

We can write it using the * notation for shorthand as:

$$\begin{tikzcd}
f^* B \arrow[r, "q(f,B)"] \arrow[d, "p(f^*B,A)"'] & B \arrow[d, "p(B,A)"] \\
A \arrow[r, "f"'] & A'
\end{tikzcd}$$

Lemma 3.21. *If $f : A \to A'$ and $A' \leq X \leq C$ in the contextual category C, then $f^*B = q(f, X)^*B$ and $q(f, B) = q(q(f, X), B)$, .i.e the following diagram commutes:*

$$\begin{tikzcd}
q(f, X)^* B \arrow[r, "q(q(f,X),B)"] \arrow[d] & B \arrow[d] \\
f^* X \arrow[r, "q(f,X)"] \arrow[d] & X \arrow[d] \\
A \arrow[r, "f"] & A'
\end{tikzcd}$$

For any contextual category C and for A an object of C we can define a contextual category C_A (similar to C/A), whose tree of objects consists of the tree of those objects which appear above A in C. More precisely, $ObC_A = \{B \in ObC | A \leq B\}$ and for $B, B' \in ObC_A$ we define $Hom_{C_A}(B, B')$ as the set of morphisms $g : B \to B'$ in C such that the following diagram commutes:

$$\begin{tikzcd}
B \arrow[r, "f"] \arrow[dr] & B \arrow[d] \\
 & A
\end{tikzcd}$$

If we have a morphism Â·$f : A \to A'$ in C, we define the contextual functor $C_f : C_{A'} \to C_A$ that maps an object B of $C_{A'}$ to f^*B in C_A and a morphism $g : B \to B'$

to f^*g, the unique morphism from f^*B to f^*B' in C_A such that $q(f, B') \circ f^*g = g \circ q(f, B)$. By condition 4a of the definition 3.19 we have that if $A \in ObC$, then $C_{id_A} = id_{C_A}$. and by condition 4b we have that if $f : A \to A'$ and $f' : A' \to A''$ in C, then $C_{f'} \circ C_f = C_{f' \circ f}$. Hence we have a functor $C_- : C^{op} \to Con$.

Definition 3.22. *If $F : C \to C'$ is a contextual functor, then if $A \in ObC$ let $F_A : C_A \to C'_{F(A)}$ be the restriction of F to C_A. Then F_A is a contextual functor and $F_- : C_- \Rightarrow C'_{F(A)}$ is a natural transformation*

$$\begin{tikzcd}
C^{op} \arrow[dd, "F"'] \arrow[dr, "C_-"] \arrow[dr, phantom, shift right=3ex, "\Downarrow F_-"] & \\
& Con \\
C'^{op} \arrow[ur, "C'_-"'] &
\end{tikzcd}$$

Lastly we will introduce Functorial Semantics, first introduced by Lawvere for algebraic theories, it gives a way of defining the concept of model of a theory U using structure preserving functors between some category $C(U)$ and a given canonical category. In the case of algebraic theories the canonical category is Set, but in clasical proposition theories the canonical category is the Boolean Algebra $\{0, 1\}$. In our situation, this category is the contextual category Fam. If U is a generalized algebraic theory, then the category of models of U is equivalent to the category with contextual functors from $C(U)$ to Fam as objects and natural transformations as morphisms. Thus we assert

$$U - alg \equiv ConFunc(C(U), Fam)$$

The inductive construction of $C(U)$ from U allows us to replace the usual inductive definition of model of U by the definition "a model of U is a contextual functor $M : C(U) \to Fam$".

Chapter 4

Calculus of Constructions

Chapter 5

Expansion Algorithm

5.1 Problem Definition

In this section we will introduce a formal definition of our problem, proposed at the beginning of the thesis, and reformulate it to an equivalent problem easier to solve.

Our problem might be formalized in the following manner: given a theory U and a derived rule of U of the form

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{\Delta \text{ is a type}}$$

we want to find an expression of U such that the rule

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n}{t : \Delta}$$

is a derived rule of U.

This problem is undecidable, in general, as it follows from the undecidability of the inhabitation in a GAT. This is why we won't try to find an algorithm that either finds a term or says that the type is empty. We aim to find an algorithm that in the case that the type is inhabited, it will find a term in a finite number of steps. In the case that the type is empty, it does not have to give an output.

We can consider this problem from the point of view of the context category $C(U)$, where it is rephrased as: given a context $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$ in $C(U)$, we want to find a morphism

$$\langle x_1, \ldots, x_n, t \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \to \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x : \Delta \rangle$$

in $C(U)$. Then if we name $A = \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle$, this is equivalent to find a morphism $\langle t_1, \ldots, t_n, t \rangle : \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n \rangle \to \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x : \Delta \rangle$ in the

category C_A, as by the following commutative diagram

$$\begin{tikzcd} A \arrow[rr, "\langle t_1,\ldots,t_n,t\rangle"] \arrow[dr, "id_A"'] & & \langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, x : \Delta\rangle \arrow[dl, "\langle x_1,\ldots,x_n\rangle"] \\ & A & \end{tikzcd}$$

for each $1 \leq i \leq n$, $x_i = t_i$. This restriction holds for all morphism $f = \langle t_1, \ldots, t_n, t_{n+1}, \ldots, t_{n+m}\rangle : B \to B'$ in C_A, by the commutativity of the diagram

$$\begin{tikzcd} B \arrow[rr, "f"] \arrow[dr] & & B' \arrow[dl] \\ & A & \end{tikzcd}$$

$t_i = x_i$ for all $1 \leq i \leq n$. That's why we will adopt the notation $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A$ to represent $\langle x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle$ and $f = \langle t_{n+1}, \ldots, t_{n+m}\rangle_A$ to represent $\langle x_1, \ldots, x_n, t_{n+1}, \ldots, t_{n+m}\rangle$ in C_A.

5.2 Expansion algorithm

In this section we will expose the expansion algorithm, that will be responsible of finding the $\langle t\rangle_A$ in C_A.

First consider G a finite subcategory of C_A such that it contains $\langle\rangle_A$, $\langle x : \Delta\rangle_A$ and $p(\langle x : \Delta\rangle_A) : \langle x : \Delta\rangle_A \to \langle\rangle_A$. The process will consist in the construction of a sequence $\{G_i\}_i$ of finite subcategories of C_A such that $G_0 = G$ and that G_{n+1} is a finite subcategory of C_A computed via the process described below from G_n and where $G_n \subseteq G_{n+1}$ for all $n \geq 0$. We say that an object/morphism of C_A is **constructible by expansion** if there exists a $n \geq 0$ such that the given object/morphism is contained in G_n. The process of constructing G_{n+1} from G_n is consists on applying the following rules to each context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A$ of G_n.

E1 For each sort symbol A with a well-formed introductory rule

$$\frac{z_1 : \Lambda_1, \ldots, z_n : \Lambda_n}{A(z_1, \ldots, z_n) \text{ is a type}}$$

such that G_n contains morphisms

$$\begin{aligned}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_1\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_2\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_2 : \Lambda_2[t_1|z_1] \rangle_A \\
&\vdots \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_w\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}] \rangle_A
\end{aligned}$$

then we add the context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, y : A(t_1, \ldots, t_w) \rangle_A$ and its projection to G_{n+1}.

E2 add the morphisms

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,y_i\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, y : \Omega_i \rangle_A$$

and their corresponding projections for each $0 \leq i \leq n$.

E3 for every operator symbol F with a well-formed introductory rule

$$\frac{z_1 : \Lambda_1, \ldots, z_n : \Lambda_n}{F(z_1, \ldots, z_n) : \Lambda}$$

if the following morphisms exist

$$\begin{aligned}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_1\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_2\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_2 : \Lambda_2[t_1|z_1] \rangle_A \\
&\vdots \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1,\ldots,y_m,t_w\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}] \rangle_A
\end{aligned}$$

then we add the morphism

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,F(t_1,\ldots,t_w)\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda[t_1|z_1, \ldots, t_w|z_w] \rangle_A$$

E4 for every morphism

$$\langle t_1, \ldots, t_{m-1} \rangle_A : \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A \to \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1} \rangle_A$$

we construct the corresponding pullback

$$\begin{tikzcd}
\langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w, y_m : \Omega_m[t_1|y_1, \ldots, t_{m-1}|y_{m-1}]\rangle_A \arrow[r, "\langle t_1,\ldots,t_{m-1},y_m\rangle_A"] \arrow[d] & \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A \arrow[d] \\
\langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w\rangle_A \arrow[r, "\langle t_1,\ldots,t_{m-1}\rangle_A"] & \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A
\end{tikzcd}$$

in G_{n+1}.

Once we have applied the expansion rules to all of the contexts, we complete G_{n+1} with the composition morphisms to ensure it is a subcategory.

Theorem 5.1. *Every object and every morphism of C_A is constructible by expansion.*

Proof. We will proof this result by structural induction over the underlying derived rules corresponding to such object (context) or morphism (operator).

Consider a context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A$ of U, which is the representative of an object in $C(U)$. This means, by the definition of context, that its underlying rule

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{\Omega_m \text{ is a type}} \tag{5.1}$$

is a derived rule of U and y_m is a variable different to $x_1, \ldots, x_n, y_1, \ldots, y_{m-1}$. Hence, the premise $\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A$ is a context and by induction it is constructible by expansion, i.e. there exists $n_0 \in \mathbb{N}$ such that $\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A \in G_{n_0}$. As (5.1) is a derived rule, it must have been derived by the only derivation rule that has a T-rule as a conclusion: CF2(a). Hence, there exists a sort symbol S with a well-formed introductory rule $\dfrac{z_1 : \Lambda_1, \ldots, z_w : \Lambda_w}{S(z_1, \ldots, z_w) \text{ is a type}}$ and derived rules

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{t_1 : \Lambda_1},$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{t_2 : \Lambda_2[t_1|z_1]},$$
$$\vdots$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{t_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}]}$$

By induction hypothesis their corresponding morphisms

$$\langle y_1, \ldots, y_{m-1}, t_j\rangle_A : \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}\rangle_A \to \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}, z_j : \Lambda[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]\rangle_A$$

are constructible by expansion, so there exist in G_{n_j} for some $n_j \in \mathbb{N}$.

Take $n = \max\{n_0, n_1, \ldots, n_j\}$. Then, all of the objects, and morphisms mentioned belong to G_n so by the expansion rule E1,

$$\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1} \rangle_A \longleftarrow \langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}, y_m : S(t_1, \ldots, t_w) \rangle_A$$

is included in G_{n+1}. And by the equivalence of intended equivalence of denotation

$$\langle y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}, y_m : S(t_1, \ldots, t_w) \rangle_A \equiv \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$$

so $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$ is constructible by expansion.

Now consider an operator

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle t_1, \ldots, t_w \rangle} \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A$$

a representative of a morphism in C_A. By induction hypothesis we know that both $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$ and $\langle z_1 : \Omega_1, \ldots, z_w : \Omega_w \rangle_A$ are constructible by expansion.

Moreover, as $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A$ is an operator, for each $1 \leq j \leq w$, the rule

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{t_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]} \tag{5.2}$$

is a derived rule of U. Now we will see that the following morphisms are constructible by expansion:

$$\begin{aligned}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1, \ldots, y_m, t_1 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1 \rangle_A \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1, \ldots, y_m, t_2 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_2 : \Lambda_2[t_1|z_1] \rangle_A \\
&\vdots \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A &\xrightarrow{\langle y_1, \ldots, y_m, t_w \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_w : \Lambda_w[t_1|z_1, \ldots, t_{w-1}|z_{w-1}] \rangle_A
\end{aligned}$$

To do this, we will use how any of the rules (5.2) are derived. To do it, we must use one of the following derivation rules: T1, CF1 or CF2(b). Hence,

- if derived by T1, we have that both

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{\Delta = \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]}$$

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{t_j : \Delta}$$

are derived rules of U. So by the first rule, due to the equivalence realization of identity of denotation, we know that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Delta \rangle_A \equiv \\ \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

hence they are the same object in C_A. And by the second one, applying induction, we have that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Delta \rangle_A$$

is constructible by expansion, so as $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Delta \rangle_A$ and $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$ are the same object in C_A, then

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

is constructible by expansion.

- if derived by CF1, we have that

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_{m-1} : \Omega_{m-1}}{\Omega_m \text{ is a type}}$$

is a derived rule of U, where y_m is a variable distinct from all $x_1, \ldots, x_n$, $y_1, \ldots, y_{m-1}$, and that there exists an i, $1 \leq i \leq m$, such that

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{t_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]} \equiv \frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Omega_m}{y_i : \Omega_i}$$

Then, by induction hypothesis, the context $\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle$ is constructible by expansion. By applying the expansion rule E2 and an argument of equivalence similar to the one above, we get that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_j \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}] \rangle_A$$

is constructible by expansion.

- if derived by CF2(b), there exists an operator symbol F with a well-formed introductory rule

$$\frac{c_1 : \Gamma_1, \ldots, c_v : \Gamma_v}{F(c_1, \ldots, c_v) : \Gamma}$$

and derived rules

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{s_1 : \Gamma_1}$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{s_2 : \Gamma_2[s_1|c_1]}$$
$$\vdots$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{s_v : \Gamma_v[s_1|c_1, \ldots, s_{v-1}|c_{v-1}]}$$

such that

$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{\Gamma[s_1|c_1, \ldots, s_v|c_v] = \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]}$$
$$\frac{x_1 : \Delta_1, \ldots, x_n : \Delta_n, y_1 : \Omega_1, \ldots, y_m : \Delta_m}{t_j = F(s_1, \ldots, s_v) : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]}$$

are derived rules of U. By the equivalence relation of identity of denotation, we have to prove that the morphism

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, F(s_1, \ldots, s_v) \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Gamma[s_1|c_1, \ldots, s_v|c_v] \rangle_A$$

is constructible by expansion. By induction we have that the following morphisms are constructible by expansion:

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, s_1 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, c_1 : \Gamma_1 \rangle_A$$
$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, s_2 \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, c_2 : \Gamma_2[s_1|c_1] \rangle_A$$
$$\vdots$$
$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, s_v \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, c_v : \Gamma_v[s_1|c_1, \ldots, s_{v-1}|c_{v-1}] \rangle_A$$

Hence, by applying expansion rule E3, we have that

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, F(s_1, \ldots, s_v) \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Gamma[s_1|c_1, \ldots, s_v|c_v] \rangle_A$$

is constructible by expansion as we wanted.

Now we will show tha the morphism

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_1, \ldots, t_w \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_w : \Lambda_w \rangle_A$$

is constructible by expansion. To do this we will prove that given a j, $2 < j \leq n$, if

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1, \ldots, y_m, t_1, \ldots, t_{j-1} \rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_{j-1} : \Lambda_{j-1} \rangle_A$$

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,t_j\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_{j-1}|z_{j-1}]\rangle_A$$

are constructible by expansion so is

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m \rangle_A \xrightarrow{\langle y_1,\ldots,y_m,t_1,\ldots,t_j\rangle_A} \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_j : \Lambda_j - 1\rangle_A$$

By expansion rule E4, the following diagram is constructible by expansion:

$$\begin{tikzcd}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_j : \Lambda_j[t_1|z_1, \ldots, t_j|z_j]\rangle_A \arrow[r, "\langle y_1,\ldots,y_m,t_1,\ldots,t_{j-1},z_j\rangle_A"] \arrow[d] & \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_j : \Lambda_j\rangle_A \arrow[d] \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle_A \arrow[r, "\langle y_1,\ldots,y_m,t_1,\ldots,t_{j-1}\rangle_A"] & \langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_{j-1} : \Lambda_{j-1}\rangle_A
\end{tikzcd}$$

and by composition we get that

$$\langle y_1, \ldots, y_m, t_1, \ldots, t_{j-1}, z_j\rangle_A \circ \langle y_1, \ldots, y_m, t_j\rangle_A = \langle y_1, \ldots, y_m, t_1, \ldots, t_j\rangle_A$$

is constructible by expansion. It remains to see that the morphsim

$$\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1, \ldots, z_w : \Lambda_w\rangle_A \xrightarrow{\langle z_1,\ldots,z_w\rangle_A} \langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w\rangle$$

is constructible by expansion. That comes directly from the fact that $\langle z_1 : \Lambda_1, \ldots, z_w : \Lambda_w\rangle_A$ is a context and the repeated application of the rule E4 as this diagram is constructible by expansion:

$$\begin{tikzcd}
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m z_1 : \Lambda_1, \ldots, z_w : \Omega_w\rangle_A \arrow[r, "\langle z_1,\ldots,z_w\rangle_A"] \arrow[d] & \langle z_1 : \Lambda_1, \ldots, z_w : \Omega_w\rangle_A \arrow[d] \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m z_1 : \Lambda_1, \ldots, z_{w-1} : \Omega_{w-1}\rangle_A \arrow[r, "\langle z_1,\ldots,z_{w-1}\rangle_A"] \arrow[d] & \langle z_1 : \Lambda_1, \ldots, z_{w-1} : \Omega_{w-1}\rangle_A \arrow[d] \\
\vdots \arrow[d] & \vdots \arrow[d] \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m, z_1 : \Lambda_1\rangle_A \arrow[r, "\langle z_1\rangle_A"] \arrow[d] & \langle z_1 : \Lambda_1\rangle_A \arrow[d] \\
\langle y_1 : \Omega_1, \ldots, y_m : \Omega_m\rangle \arrow[r, "\langle\rangle_A"] & \langle\rangle_A
\end{tikzcd}$$

□

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